

AVNI SINGH

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SUMMARY

Software Engineer with 1+ years of hands-on experience building production-grade backend systems, distributed applications, and scalable software. Experienced in developing REST APIs, full-stack applications, and distributed workflows using Python, JavaScript, FastAPI, React, PostgreSQL, and MongoDB, with a strong foundation in software engineering and modern AI-powered applications.

EXPERIENCE

AI/ML Intern | Bharat Forge

Pune, On-Site | Jul 2025 – Jul 2026

- Fleet Management System

[GitHub](#)

- Designed and developed **backend services and REST APIs** for a production fleet management platform, enabling mission creation, task dispatch, lifecycle management, and real-time communication between warehouse applications and autonomous systems
- Built **graph-based task scheduling** and **conflict management** modules with dynamic re-routing, while implementing request validation, structured logging, and persistent mission state using **MongoDB, MQTT, and Express**
- Developed real-time operational dashboards for fleet health, mission execution, and dispatch analytics using **React, TanStack Query, and Recharts**, providing production teams with live system visibility and performance insights

- Warehouse Management System

[GitHub](#)

- Developed a **full-stack warehouse workflow platform** enabling QR-based task creation, automated transport dispatch, and end-to-end task lifecycle management using **React, Node.js, Express, and PostgreSQL**
- Designed and implemented **backend APIs** integrating the Warehouse Management System with the Fleet Management System, enabling reliable task dispatch, callback-based status synchronization, and consistent workflow tracking across distributed services
- Built transport monitoring dashboards with live task status, robot statistics, historical snapshots, and system health monitoring, improving operational visibility through persistent event logging and real-time data synchronization

Research Intern | DRDO (Defence Research and Development Organisation)

Mussoorie, On-Site | Dec 2024 – Jan 2025

- Conducted a comparative study of AI/ML-enabled air defence systems, analyzing how computer vision, sensor fusion, predictive analytics, and autonomous decision-making improve threat detection, target classification, multi-target tracking, and real-time decision-making, strengthening my understanding of real-world AI applications in safety-critical environments

PROJECTS

Interview Coach AI | LangGraph, LangChain, FastAPI, ChromaDB, MCP, Python

Jul 2026 – Ongoing

- Designing a multi-agent AI mock interview system using **LangGraph**, where interviewer, evaluator, and router **agents** run in a stateful decision loop that adapts question difficulty and topic based on live candidate performance.
- Building a **RAG** pipeline over a **vector database** (ChromaDB) of role-specific interview questions, using sentence-transformer embeddings for semantic retrieval, and integrating an **MCP** server for real-time web-search tool access.
- Implementing prompt-based **guardrails** to prevent answer leakage and topic drift, paired with an LLM-as-judge **evals** pipeline that scores response quality and retrieval relevance against a fixed test set.

DeepSea Vision | Computer Vision, Deep Learning, FastAPI

Mar 2025 – May 2025

- Built a deep learning image enhancement pipeline using a custom **SRGAN** architecture to restore degraded underwater images and improve visual quality for downstream computer vision tasks
- Implemented adversarial and perceptual learning using pretrained **VGG-19** features, understanding how feature representations improve visual realism beyond pixel-level reconstruction
- Trained and evaluated the model on paired underwater datasets, optimizing image enhancement performance and analyzing its impact on downstream vision applications

PlotPoint | GPT-2, LoRA, RAG, FAISS, Stable Diffusion, Gemini-Pro

Aug 2024 – Oct 2024

- Built an AI pipeline for automated movie script and scene generation by combining Retrieval-Augmented Generation (RAG), GPT-2 fine-tuning, and Stable Diffusion for context-aware text and image generation.
- Developed a semantic retrieval pipeline using text **chunking**, sentence-transformer **embeddings**, and FAISS vector search to provide relevant context to Gemini-Pro, improving script coherence and reducing hallucinations.
- Evaluated multiple language models and generation strategies using Cosine Similarity, Perplexity, and CLIP Score to compare script relevance, language quality, and text-to-image alignment.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL

Machine Learning: Scikit-learn, PyTorch, Transformers, Hugging Face, Feature Engineering, Model Evaluation, Cross Validation, Hyperparameter Tuning, Classification, Regression, OpenCV, YOLO, GANs, Diffusion Models

Data Analysis: Pandas, NumPy, Data Preprocessing, Exploratory Data Analysis (EDA), Data Visualization

Applied AI: Large Language Models (LLMs), AI Agents, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Model Context Protocol (MCP), Prompt Engineering, LLM Evals, Guardrails, Generative AI

Backend & AI Infrastructure: FastAPI, REST APIs, Model Inference, FAISS, ChromaDB, Vector Databases

Databases & Tools: PostgreSQL, MongoDB, Git, Docker, Linux

Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, System Design, Concurrency

Cloud: AWS

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning | [Symbiosis Institute of Technology, Pune](#) Aug 2022 – May 2026
CGPA: 7.8

ACHIEVEMENTS

- Selected as one of the **7 winners** from **7,800+ applicants** nationwide in the **Kalyani Group & WeCP Data, AI & Robotics Hackathon 2025**, earning selection for an AI/ML internship at Bharat Forge through a multi-stage technical evaluation. [Announcement](#)

CERTIFICATIONS AND PUBLICATIONS

- [ChatGPT for Educators & Researchers & Professionals](#) – World Technocon
- [GitHub Copilot Global Bootcamp](#) – Microsoft Learn Student Ambassadors
- [What Is Generative AI?](#) – LinkedIn
- [Introduction to Generative AI](#) – Simplilearn
- **Publication:** LLM-based Abstractive Legal Document Summarization using Transformer Architectures – ICIRD 2024, Shinawatra University